THE TENSOR-INTERNAL-HOM DIAGONAL MORPHISM K TENSOR RHOM(M, L) -> RHOM(M, K TENSOR L)

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Allowed standard prerequisites (editorial).

- For K-flat K and complexes M, I there is a canonical map Tot(K tensor Hom(M, I)) -> Hom(M, Tot(K tensor I)). invoked at proof line 18826. Supplies the first map of the composite defining the morphism.

Source reference. Original lemma 0BYM: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18036–18036. Cited at proof line 18826 as the first arrow of the composite defining the morphism; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Given complexes K, L, M in $D(R)$ there is a canonical morphism*

$$K \otimes^{\mathbf{L}}_R R\mathrm{Hom}_R(M, L) \longrightarrow R\mathrm{Hom}_R(M, K \otimes^{\mathbf{L}}_R L)$$

in $D(R)$ functorial in K, L, M.

Proof. Choose a K-flat complex $K^\bullet$ representing K, and a K-injective complex $I^\bullet$ representing L, and choose any complex $M^\bullet$ representing M. Choose a quasi-isomorphism $\mathrm{Tot}(K^\bullet \otimes_R I^\bullet) \to J^\bullet$ where $J^\bullet$ is K-injective. Then we use the map

$$\mathrm{Tot}\left(K^\bullet \otimes_R \mathrm{Hom}^\bullet(M^\bullet, I^\bullet)\right) \to \mathrm{Hom}^\bullet(M^\bullet, \mathrm{Tot}(K^\bullet \otimes_R I^\bullet)) \to \mathrm{Hom}^\bullet(M^\bullet, J^\bullet)$$

where the first map is the map from Lemma 0BYM. □

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